

Goals:

- ✓ Generic and flexible package for auxiliary field Monte Carlo simulations of fermion-boson lattice models
- ✓ Efficient implementation on modern HPC systems
- ✓ Benchmarking and golden standards

Dirac systems

$$\hat{H} = -t \sum_{\langle ij \rangle} (\hat{c}_i^\dagger \hat{c}_j + \hat{c}_j^\dagger \hat{c}_i) - \lambda \sum_{\mathbf{Q}} \left( \sum_{\langle (i,j) \rangle \in \mathbf{Q}} \nu_{i,j} i (\hat{c}_i^\dagger \sigma \hat{c}_j - \hat{c}_j^\dagger \sigma \hat{c}_i) \right)^2 - U \sum_i (\hat{n}_i - 1)^2$$

Y. Liu, Z. Wang, T. Sato, M. Hohenadler, C. Wang, Wenan Guo, and F. F. Assaad, Nature Communications 10 (2019), no. 1, 2658, PRL 126 (2021), 205701, arXiv:2212.11395

**Bandwidth-controlled transitions**

$\pi$ -Rotation QSH to SSC

**Doping induced transitions**

Skyrmion superconductivity

**Candidate theory for DQCP**

$$S = \int d^2x d\tau \frac{1}{G} (\partial_\mu \hat{\varphi}(x, \tau))^2 + 2\pi i \Gamma [\hat{\varphi}] \quad \Gamma[\hat{\varphi}] = \frac{\epsilon_{abcde}}{\text{Area}(S^4)} \int_0^1 du \int d^2x d\tau \hat{\varphi}_a \partial_x \hat{\varphi}_b \partial_y \hat{\varphi}_c \partial_\tau \hat{\varphi}_d \partial_u \hat{\varphi}_e$$

Z. Wang, M. P. Zaletel, R. S. K. Mong, and F. F. Assaad, PRL 126 (2021), 045701.

See also: J. S. Hofmann, E. Khalaf, A. Vishwanath, E. Berg, and J.-Y. Lee, PRX 12 (2022), 011061, for TBG.

SU(N) Kondo-Heisenberg

$$\hat{H} = -t \sum_{\langle ij \rangle} (\hat{c}_i^\dagger \hat{c}_j + h.c.) + \frac{J_k}{2} \sum_{\mathbf{r}} \hat{c}_{\mathbf{r}}^\dagger \sigma \hat{c}_{\mathbf{r}} \cdot \hat{S}_{\mathbf{r}} + J_h \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \hat{S}_{\mathbf{r}} \cdot \hat{S}_{\mathbf{r}'}$$

**Kondo breakdown transition**

B. Danu, M. Vojta, F. F. Assaad, and T. Grover, PRL 125 (2020), 206602

**Dissipation induced long range order**

QCP between antiferromagnetic heavy fermion metal and heavy fermion metal

B. Danu, M. Vojta, T. Grover, and F. F. Assaad, PRB 106 (2022), L161103.

**Realization of marginal Fermi liquid at QCP**

between antiferromagnetic heavy fermion metal and heavy fermion metal

Z.H Liu, B. Frank, L. Janssen, M. Vojta, and F. F. Assaad, PRB 107 (2023), 165104, arXiv:2212.11283

See also:  
B. Danu, Z. Liu, F. F. Assaad, and M. Raczkowski, PRB 104 (2021), 155128.  
M. Raczkowski and F. F. Assaad, PRR 2 (2020), 013276.

The ALF package

**Kinetic**

$$\hat{H} = \sum_{k=1}^{M_T} \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y} \hat{c}_{x\sigma s}^\dagger T_{xy}^{(ks)} \hat{c}_{y\sigma s} + \sum_{k=1}^{M_V} U_k \left\{ \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \left[ \left( \sum_{x,y} \hat{c}_{x\sigma s}^\dagger V_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) + \alpha_{ks} \right] \right\}^2 + \sum_{k=1}^{M_I} \hat{Z}_k \left( \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y} \hat{c}_{x\sigma s}^\dagger I_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) + \hat{H}_{\text{Ising}}$$

**Potential (sum of perfect squares)**

**Coupling of fermions to Ising field**

- Model can be specified at minimal programming cost
- MPI/OpenMP implementation
- Fortran 2008 standard
- SU(N<sub>col</sub>) symmetric in colors  $\sigma$ -index
- Arbitrary Bravais lattice for d=1, 2
- Block diagonal in flavors, N<sub>fl</sub> s-index
- Projective and finite T approaches

Wissenschaftliche Literaturversorgungs- und Informationssysteme (LIS)

Center of excellence – complexity and topology in quantum matter

Electron-phonon/Lattice gauge

**Exotic phases and phase transitions in an SSH model on a square lattice**

A. Götz, S. Beyl, M. Hohenadler, and F. F. Assaad, PRB 105 (2022), 085151, arXiv:2307.07613

$$\hat{H} = \sum_{\langle ij \rangle} (-t + g \hat{X}_{(i,j)}) \sum_{\sigma=1}^N (\hat{c}_{i,\sigma}^\dagger \hat{c}_{j,\sigma} + h.c.) + \sum_{\langle ij \rangle} \left[ \frac{\hat{P}_{(i,j)}^2}{2m} + \frac{k}{2} \hat{X}_{(i,j)}^2 \right] - \lambda \sum_{\langle ij \rangle} \left( \sum_{\sigma=1}^N \hat{c}_{i,\sigma}^\dagger \hat{c}_{j,\sigma} + h.c. \right)^2$$

**Emergent Dirac fermions**

**Flux/plaquette**

**Instabilities of nested FS**

At t=0 model has same symmetries as Z<sub>2</sub> lattice gauge theory

$$\hat{H} = \sum_{\langle ij \rangle} \sigma_{(i,j)}^z \sum_{\sigma=1}^N (\hat{c}_{i,\sigma}^\dagger \hat{c}_{j,\sigma} + h.c.) - h \sum_{\langle ij \rangle} \sigma_{(i,j)}^x$$

F. F. Assaad and Tarun Grover, PRX 6 (2016), 041049  
S. Gazit, M. Randeria, and A. Vishwanath, Nat Phys 13 (2017), 484  
S. Gazit, F. F. Assaad, S. Sachdev, A. Vishwanath, and C. Wang, PNAS (2018)

Frustrated spin systems

$$\hat{H} = 2K \sum_{i \in A, \delta} \hat{S}_i^\delta \hat{S}_{i+\delta}^\delta + J \sum_{i \in A, \delta} \hat{S}_i \cdot \hat{S}_{i+\delta}$$

**Kitaev-Heisenberg**

T. Sato and F. F. Assaad, PRB 104 (2021), L081106

**Simulating spins with fermions.**

$$\hat{S}_i^\delta = \frac{1}{2} \sum_{s,s'} \hat{f}_{i,s}^\dagger \sigma_{s,s'}^\delta \hat{f}_{i,s'}$$
$$\sum_s \hat{f}_{i,s}^\dagger \hat{f}_{i,s} \equiv \hat{n}_i = 1$$
$$\hat{H}_{QMC} = |K| \sum_{i \in A, \delta} \left( \hat{S}_i^\delta + \frac{K}{|K|} \hat{S}_{i+\delta}^\delta \right)^2 - \frac{J}{8} \sum_{i \in A, \delta} \left( [\hat{D}_{i,\delta}^\dagger + \hat{D}_{i,\delta}]^2 + [i\hat{D}_{i,\delta} - i\hat{D}_{i,\delta}^\dagger]^2 \right) + U \sum_i (\hat{n}_i - 1)^2$$
$$\hat{D}_{i,\delta}^\dagger = \sum_s \hat{f}_{i,s}^\dagger \hat{f}_{i+\delta,s}$$

Constraint commutes with Hamiltonian dynamics  $[\hat{H}_{QMC}, \hat{n}_i] = 0$

Phase of fermion determinant is pinned to  $\pi\pi$

**RuCl<sub>3</sub>**

$$\hat{H}(\varphi) = \sum_{\langle ij \rangle} \left[ K \hat{S}_i^\gamma \hat{S}_j^\gamma + \Gamma \hat{S}_i^\alpha \hat{S}_j^\beta + J_1 \hat{S}_i \cdot \hat{S}_j \right] + J_3 \sum_{\langle\langle ij \rangle\rangle} \hat{S}_i \cdot \hat{S}_j - \mu_B \sum_i \mathbf{B}(\varphi) \cdot \mathbf{g} \hat{S}_i$$

$(J_1, J_3, K, \Gamma) = (-0.5, 0.5, -5.0, 2.5)$  [meV]  $\mathbf{g} = \text{diag}[2.3, 2.3, 1.3]$  Winter et al. Nat. Comm. 8 (2017), PRL 120 (2018)

QMC

$k = \frac{\partial^2 F(\varphi)}{\partial \varphi^2} \Big|_{\varphi=0}$

(Work in progress)

Scale-invariant magnetic anisotropy in RuCl<sub>3</sub> at high magnetic fields

Scale-invariant magnetic anisotropy in RuCl<sub>3</sub> at high magnetic fields

Scale-invariant magnetic anisotropy in RuCl<sub>3</sub> at high magnetic fields